SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

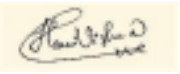
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
Student Name:	HARI VISHNU N	**Reg. No.:**	192524319
Section / Batch:	SLOT D / 2025	**Date:**	10-10-2026

Code of Conduct

I HARI VISHNU N / 192524319 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool. • Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**

8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**
9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Annexure A – Error Analysis Task

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error).

Statement	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	(1) "judges a machine to be intelligent if its internal reasoning always follows the laws of logic" – wrong criterion. (2) "the standard example of the 'thinking rationally' approach" – wrong category (it is 'acting humanly').
S2	Incorrect	(1) "AI winters ... occurred in the 1950s, before the 1956 Dartmouth workshop" – the winters came after 1956 (mid-1970s and late 1980s–early 1990s). (2) "That workshop then revived the field" – Dartmouth launched/founded the field; there was nothing to revive. ("coined the term 'artificial intelligence'" is

		correct.)
S3	Incorrect	"A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance" (rationality confused with omniscience), and the consequence drawn from it: "Hence ... [the taxi] was not rational".
S4	Incorrect	(1) "Performance measure: steering, accelerator, brake, horn" – these are actuators. (2) "Actuators: safe, fast, legal and comfortable trip" – this is the performance measure. (Environment and Sensors entries are correct.)
S5	Incorrect	(1) "partially observable" – chess is fully observable. (2) "stochastic" – chess is deterministic. (3) "episodic" – chess is sequential. ("static" and "discrete" are correct.)

Statement	Correct / Incorrect	Erroneous phrase / claim
S6	Correct	None – the statement is accurate.

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)
S1 – criterion of the Turing Test: "internal reasoning always follows the laws of logic"	The Turing Test is an operational, behavioural test: an interrogator converses in writing and the machine passes if its replies cannot be told apart from a human's. It never inspects internal reasoning. The engineer confused it with the 'laws of thought' (logic-based correct reasoning) idea.
S1 – classification as "thinking rationally"	Confusion among the four approaches (thinking humanly, thinking rationally, acting humanly, acting rationally). A test based on human-like external behaviour is the standard example of 'acting humanly'; 'thinking rationally' is the logicist tradition (Aristotle's syllogisms, laws of thought).
S2 – AI winters "in the 1950s, before 1956"	Reversed chronology of the eras of AI. A field must exist and over-promise before it can suffer a winter; the winters (first mid-1970s, second late 1980s–early 1990s) followed unmet expectations, e.g. limits of perceptrons, machine translation failures and the collapse of the expert-systems market.
S2 – Dartmouth workshop "revived" the field	Misunderstanding of the significance of the 1956 Dartmouth workshop (McCarthy, Minsky, Shannon, Rochester): it was the birth of AI as a named discipline, not a revival. Later revivals came from expert systems (1980s) and machine learning / deep learning (2010s).
S3 – rational agent "always achieves the best possible outcome ... knows the actual result" (and so the taxi 'was not rational')	Rationality is confused with omniscience (perfection). A rational agent maximises EXPECTED performance given its percept sequence and knowledge; it cannot know actual outcomes in advance. Judging the decision by an unforeseeable bad outcome (landslide) instead of by the information available when the decision was made led to the wrong verdict.
S4 – Performance measure filled with steering, accelerator, brake, horn	Misunderstanding of the P in PEAS: the performance measure is the criterion by which the agent's behaviour is judged (safety, speed, legality, comfort, profit), not the controls the agent uses.

S4 – Actuators filled with "safe, fast, legal and comfortable trip"	Misunderstanding of the A in PEAS: actuators are the devices through which the agent acts on the environment (steering, accelerator, brake, horn/signals, display). The P and A entries were swapped – 'how the agent acts' was confused with 'how it is judged'.
S5 – "partially observable"	In chess every percept shows the whole board and all pieces, so it is fully observable. The student confused hidden opponent intentions or the large search space with partial observability.
S5 – "stochastic"	Chess has no chance element: the next state is completely determined by the current state and the action (deterministic). The opponent's unpredictability is a multi-agent (competitive) issue, not stochasticity.
S5 – "episodic"	Each move affects all later positions, so decisions are not independent episodes; the task is sequential. Separate games being independent was wrongly applied to the moves inside a game.

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write "No change" and justify why.

Statement	Corrected statement	Justification
S1	The Turing Test (1950) judges a machine to be intelligent if a human interrogator, after asking written questions, cannot tell whether the replies come from a person or a machine. It is therefore the standard example of the "acting humanly" approach to AI.	The test is purely behavioural: only external behaviour is judged, never internal reasoning. Passing it needs NLP, knowledge representation, automated reasoning and machine learning – a human-like-behaviour (acting humanly) goal. 'Thinking rationally' concerns correct inference by laws of logic.
S2	The first AI winter (a period of reduced funding and interest) occurred in the mid-1970s and a second in the late 1980s– early 1990s, i.e. after the 1956 Dartmouth workshop. That workshop coined the term "artificial intelligence" and launched AI as a field.	Dartmouth (1956) marks the birth of AI. Winters are reactions to over-promising and therefore can only occur after the field has started.
S3	A rational agent chooses the action expected to maximise its performance measure, given its percept sequence to date and its built-in knowledge; it does not need to know actual outcomes in advance (rationality is not omniscience). Hence a taxi that picks a route later blocked by an unforeseeable landslide was still rational.	Rationality is defined by expected, not actual, performance. Omniscience is impossible in practice. The landslide was not in the taxi's percepts or knowledge, so the decision remains rational; only an outcome-based (hindsight) judgement calls it irrational.
S4	Performance measure: safe, fast, legal and comfortable trip (and maximise profit). Environment: roads, other traffic, pedestrians, weather. Actuators: steering, accelerator, brake, horn. Sensors: cameras, GPS, speedometer.	P and A were swapped. P = criteria for success, A = means of acting. The Environment and Sensors entries were already correct and are unchanged.
S5	Chess played without a clock is a fully observable, deterministic, sequential, static and discrete task environment (multi-agent and competitive).	The whole board is visible (fully observable); a move has exactly one outcome (deterministic); moves affect future positions (sequential); with no clock the world does not change while the agent deliberates (static); finite squares/pieces/moves (discrete). The opponent makes it multi-agent, not stochastic.

S6	No change.	The statement is correct: a model-based reflex agent maintains internal state, updated with a transition model (how the world evolves on its own and how the agent's actions change it). This lets it handle partial observability, e.g. remember a car that has moved into its blind spot.

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road.

(i) Using the four factors that decide whether an agent is rational, argue whether the taxi's choice was rational.

(ii) Name the properties of the taxi's task environment that make perfect foresight impossible.

(iii) Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

(i)	The four factors that decide rationality: (1) the performance measure defining success, (2) the agent's prior knowledge of the environment, (3) the actions it can perform, (4) its percept sequence to date. Applying them: (1) PM = safe, fast, legal, comfortable trip; (2) prior knowledge = road map and normal highway behaviour; (3) actions = it may choose any route, steer, brake, re-route; (4) percept sequence = every available traffic report showed the highway clear. Given exactly this information, the highway was the action expected to maximise the performance measure, so the choice was rational. The landslide was in neither the percepts nor the prior knowledge, i.e. unforeseeable. Rationality maximises expected, not actual, performance, so bad luck does not make the decision irrational (the taxi would have been irrational only if it had ignored the reports, e.g. picked a road reported blocked). After the landslide, rational behaviour is to stop safely and re-plan another route.
(ii)	Partially observable (it cannot sense geological or road conditions ahead), stochastic (landslides, accidents and other drivers' behaviour are not fully determined by the current state and action) and dynamic (traffic, weather and road conditions change while the taxi drives and deliberates). Also sequential, continuous and multi-agent. Partial observability + stochasticity (+ dynamics) make perfect foresight impossible.
(iii)	A simple reflex taxi acts only on the current percept through condition–action rules. Example: a car in the next lane drops into the taxi's blind spot; the current percept shows an empty lane, so the rule 'lane clear → change lane' fires and causes a collision (similarly it cannot tell from one frame whether the car ahead is braking). A model-based reflex taxi keeps an internal state: the recently observed positions and speeds of nearby vehicles, a model of how the world evolves (vehicles keep moving at similar speeds) and of how its own actions change the world (steering/lane change changes its position). Updating this state with each percept lets it infer the car is still beside it and avoid the manoeuvre.

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the **Missionaries and Cannibals** problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move

everyone to the right bank using the **fewest crossings**.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation	Incorrect	The boat's position is missing. (M, C) does not say which bank the boat is on, so states with the same (M, C) but different boat sides (different successors) are merged and applicable actions cannot be determined. Needs (M, C, B).
Initial state	Incorrect	(3, 3) does not record that the boat is on the left bank. Should be (3, 3, L).
Actions	Correct	The five loads (1M, 2M, 1C, 2C, 1M+1C) are the complete set of non-empty loads of 1–2 people. (They are applicable only if enough people are on the boat's bank, which requires the boat side in the state – see State representation.)
Goal test	Incorrect	M = C is true in the initial state (3,3) and in (2,2), (1,1), (0,0). The goal is that everyone is on the right bank: (0, 0, R).
Legality check	Incorrect	Only the left bank is checked. The right bank must also be safe, e.g. (1, 0): left 1M, 0C passes, but the right bank has 2M and 3C – illegal.
Path cost	Incorrect	Counts the people carried. The objective is fewest crossings, so each crossing must cost 1.
Note N1	Incorrect	Wrong order. It must be: goal formulation → problem formulation → search → execution.

Component / Note	Correct / Incorrect	What is wrong
Note N2	Incorrect	Without the boat, 4×4 = 16 (M, C) pairs of which 10 are legal. With the boat there are 4×4×2 = 32 states, of which 20 are legal (16 of them reachable from the start).

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error (component / note)	Root cause
State representation	Violates the principle that a state must contain all information needed to decide which actions apply and what they do (and to test the goal). The boat location was abstracted away although it determines the direction of the crossing and which bank people are taken from.
Initial state	A direct consequence of the incomplete state description; the problem statement says the boat starts on the left bank but the formulation cannot express it.
Goal test	The goal test must be true only for goal states. The student tested a property (balance of M and C, which is a safety/legality idea) instead of the stated objective (everyone on the right bank).
Legality check	The safety constraint applies to BOTH banks. The student derived the constraint for one bank only and forgot that the right bank counts are (3−M, 3−C).
Path cost	Path cost must encode the performance measure. The objective is fewest crossings, so the step cost is 1 per crossing; the student optimised a different quantity (people carried).
Note N1	Violates the problem-solving-agent cycle: the goal must be formulated first, then the problem (states/actions to consider), and only then can search run; search cannot start on an undefined problem.
Note N2	The state space was counted over the incomplete (M, C) representation, so the boat side (factor 2) was ignored; counts of total and legal states must follow the correct state representation.

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	Corrected formulation	Justification
State representation	(M, C, B): M, C = missionaries and cannibals on the LEFT bank (0–3 each); B ∈ {L, R} = bank where the boat is. Right-bank counts are (3−M, 3−C).	The boat side fixes the direction of every crossing and which bank people can be taken from, so the state is complete.
Initial state	(3, 3, L)	Everyone and the boat are on the left bank.
Goal test	(M, C, B) = (0, 0, R)	Only this state has everyone (and necessarily the boat) on the right bank; M = C is true in non goal states such as (3,3).
Legality check	State is legal iff both banks are safe: (M = 0 or C ≤ M) AND (M = 3 or C ≥ M).	The second part is the right-bank test: 3−M = 0 or 3−C ≤ 3−M, i.e. M = 3 or C ≥ M. The missionaries must not be outnumbered on either bank.
Path cost	Step cost = 1 per crossing; path cost = number of crossings.	Objective = fewest crossings, so each crossing costs the same regardless of load.
Note N1	Goal formulation → problem formulation → search → execution.	The goal defines what is wanted, the problem formulation decides states and actions, search then finds the action sequence, which is finally executed.

Component / Note	Corrected formulation	Justification

Note N2	32 states (4 × 4 × 2), of which 20 are legal (the 10 legal (M, C) pairs for each boat side); 16 are reachable from (3, 3, L).	Legal (M, C) pairs: (0,0),(0,1),(0,2),(0,3),(1,1),(2,2),(3,0),(3,1),(3,2), (3,3). Doubling for the boat side gives 20 legal states of 32.

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M	(2, 3, R)	Illegal – left bank has 2M and 3C, so cannibals outnumber missionaries.
2M	(1, 3, R)	Illegal – left bank has 1M and 3C (cannibals outnumber missionaries).
1C	(3, 2, R)	Legal – left 3M, 2C is safe; right 0M, 1C is safe (no missionaries).
2C	(3, 1, R)	Legal – left 3M, 1C is safe; right 0M, 2C is safe (no missionaries).
1M + 1C	(2, 2, R)	Legal – left 2M, 2C is safe; right 1M, 1C is safe.

(ii) Give one state that the student's legality check accepts although it is actually illegal.

(iii) Using your corrected path cost, state the cost of an optimal solution.

(ii)	State (1, 0) (boat on either bank), e.g. (1, 0, R). The student's check C ≤ M on the left bank is satisfied (0 ≤ 1), but the right bank then holds 2 missionaries and 3 cannibals, so the missionaries are outnumbered and the state is actually illegal. Another example: (2, 1) – left 2M, 1C passes, but the right bank has 1M, 2C.
(iii)	The optimal solution has 11 crossings, so the path cost is 11 (found by breadth-first search over the 16 reachable legal states). One optimal solution: 1M+1C across → 1M back → 2C across → 1C back → 2M across → 1M+1C back → 2M across → 1C back → 2C across → 1C back → 2C across.

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. **Conventions:** graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. **BFS:** do not add a node that is already explored or already in the queue. **DFS:** use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. **UCS:** write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)

5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1	BFS, iteration 3 (queue after expansion)	Queue given as [B, E, G, A]. A was enqueued in iteration 2, so in a FIFO queue it stays ahead of the newly added B, E, G: correct queue [A, B, E, G]. (Carried forward and counted once: iteration 4 dequeues B instead of A, A is missing from the explored set, the later queues, and "6 nodes removed" – correct is 7.)
2	DFS, iteration 1 (stack after pushing children)	Stack [D, C] puts D on top, so D is popped first. The alphabetically first child C must be on top: [C, D]. (The same wrong push order is repeated in iteration 2, [G, E, B, C], and gives the wrong path S–D–G, cost 9, 3 nodes – carried forward, counted once. Correct: S–C–A–B–G, cost 14, 5 nodes.)
3	UCS, iteration 4 (expanding E, g = 5)	G is reachable through E at cost 5 + 3 = 8 < 9, but the entry G:9(D) was not replaced. Correct queue: B:7(D), G:8(E), A:9(C). (Carried forward: A expanded in iteration 6, G removed at g = 9, cost 9 and 7 removals – correct is cost 8 via S–D–E–G, 6 removals.)
4	Remark R1	Wrong. An explored set IS needed in BFS graph search: the graph is undirected with several paths, so a node can be generated again (e.g. S is a neighbour of C and D). BFS 'does not revisit' only because explored/frontier checks stop it. The convention also requires the column.
5	Remark R2 (first claim: all three optimal)	False. Only UCS is optimal for non-uniform edge costs. BFS returns the shallowest path (S–D–G, cost 9 > 8); DFS returns S–C–A–B–G with cost 14.

No.	Where (trace / iteration / remark)	What is wrong
6	Remark R2 (second claim: traces confirmed correct)	Invalid verification. All three traces contain errors; identical final answers do not validate the steps (the agreement is itself false: DFS = 14, UCS = 8). Each trace must be checked iteration by iteration against its rules.

7	Remark R3	Wrong. Comparing only the number of nodes removed ignores solution quality. The correct DFS count is 5 (not 3); it is still the fewest, but its path (cost 14) is the worst, whereas UCS is optimal with 6 removals.

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	Violates the FIFO discipline of BFS: nodes leave in the order they entered. A, inserted earlier, was placed after the later nodes B, E, G. This made B be dequeued before A and so changed the explored set, queues and the node count.
2	Violates the stated push rule (alphabetically first child must be popped first, i.e. be on top). Pushing the children in the wrong order sent DFS down the S–D branch instead of S–C, so the path, cost and count are all wrong.
3	Violates the UCS update rule: replace a frontier entry whenever a cheaper path is found (relaxation). Missing G:8(E) kept the wrong priority order, so A (g = 9) was expanded before G and the wrong cost (9) was reported.
4	Confuses tree search with graph search. In graph search, cycles and multiple paths exist (undirected edges lead back to S), so an explored set is required to avoid re-expanding or re-adding nodes (infinite loops in the worst case).
5	Confuses 'a path found' with 'the optimal path'. BFS is optimal only for equal step costs; DFS is not optimal; only UCS guarantees least cost with different edge costs. The student also never computed the other paths (cost 14, cost 8).
6	Common-mode error: using agreement among traces as proof of correctness. Wrong traces can end at the same answer; correctness comes from applying the stated conventions at every iteration.
7	Uses a single, wrong metric (nodes removed) to rank strategies and relies on a wrong count. Strategies must be compared on completeness, optimality (path cost) and time/space together; fewer removals does not mean best.

(c) Write the corrected traces and justify each correction.

(i) BFS: the corrected queue after iteration 3 and the full order in which nodes are dequeued.

Queue after iteration 3	[A, B, E, G]
Order of dequeued nodes	S, C, D, A, B, E, G (G is the goal when removed at iteration 7)

(ii) DFS: the corrected trace.

Iteration	Node Popped	Stack after Pushing Children
1	S	[C, D]
2	C	[A, D]
3	A	[B, D]

Iteration	Node Popped	Stack after Pushing Children

4	B	[G, D]
5	G (goal)	[D]

(iii) UCS: the corrected trace.

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C)
5	B	7	G:8(E), A:9(C)
6	G (goal)	8	A:9(C) – search stops, goal found

Justification of the corrections	BFS: the queue is FIFO. A entered at iteration 2 (child of C); B, E, G entered at iteration 3 (children of D; S is explored). So after iteration 3 the queue is [A, B, E, G] and A is dequeued at iteration 4. Its neighbours B (already in queue) and C (explored) add nothing; B's neighbours A, D (explored) and G (in queue) add nothing, etc. G is removed at iteration 7 (7 removals); G was generated from D, so the path is S–D–G, cost 9. DFS: the first child to be popped must be on top, so push C then D → [C, D]. DFS then goes deep: C → A → B → G. In iteration 4, D is not pushed again from B because it is already on the stack. G is popped at iteration 5: path S–C–A–B–G, cost 3+6+2+3 = 14, 5 removals. UCS: always expand the lowest g (ties alphabetical) and replace an entry when a cheaper path appears. Expanding E gives G = 5+3 = 8 < 9, so G:8(E) replaces G:9(D) and is removed before A:9. Expanding B gives A = 9 (not cheaper) and G = 10 (not cheaper), so no change. G is removed at iteration 6 with g = 8: path S–D–E–G, cost 8, 6 removals.

(d) Apply the corrected traces.

(i) Complete the comparison table.

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS	S–D–G	9	7	No – fewest edges (2), but cost 9 > 8; BFS is optimal only for equal step costs
DFS	S–C–A–B–G	14	5	No – the most expensive path
UCS	S–D–E–G	8	6	Yes – uniform-cost search always returns the

				least-cost path

(ii) Use iteration 3 of the corrected BFS trace to explain why an explored set is needed in graph search.

(ii)	In corrected iteration 3, D is dequeued and its neighbours are B, E, G and S. S was already expanded in iteration 1 and is in the explored set, so it must NOT be added again. Without the explored set the queue would become [A, B, E, G, S]; S would be re-expanded, would re-add C and D (undirected edges lead back), and the search would loop and repeat work. The frontier check also stops duplicates (e.g. B, already in the queue when A is expanded). Hence BFS graph search needs the explored set (and the frontier check).

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)

0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.
C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where (task / step / conclusion)	What is wrong
1	Task A, step 3 (Suck)	Belief state left as {S6, S8}. Suck must be applied to every state: S6 (R,C,D) → S8 (R,C,C) and S8 → S8, giving {S8}.
2	Task B, step 0 (after percept)	{S1, S2, S3, S6} contains S2 and S6, in which the agent is in the RIGHT square. [Left, Dirty] is consistent only with agent in Left and Left square dirty: {S1, S3}. (Carried forward and counted once: step 1 predicted {S4, S5, S7, S8} should be {S5, S7}.)
3	Task B, step 2 (predicted)	{S6, S7}: Right applied to S7 (L,C,C) must move the agent to S8 (R,C,C). Correct predicted belief state: {S6, S8}.
4	Conclusion C1	Wrong. Sensorless problems can be solved: after [Suck, Right, Suck] the belief state is {S8}, a goal state, so the plan is a solution. (The 'two states' premise was only carried from error 1; also the location is not uncertain – S6 and S8 are both Right.)
5	Conclusion C2	{S6} = (R, C, D): the agent knows the state, but the Right square is still dirty, so the goal is NOT reached; a further Suck is needed (→ S8).
6	Conclusion C3	Belief states are SUBSETS of the physical states, so there are 2^8 = 256 belief states, not 8; searching belief states is much harder than searching physical states.
7	—	No further independent errors. Task A steps 1–2 and the Task B percept updates at steps 1–2 are correct; the Task B step 1 predicted set is only carried forward from No. 2.

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)

1	Prediction step misapplied: the belief state after an action is {RESULT(s, a) : s in b}, with the action applied to EVERY state. The student treated Suck as having no effect on S6, although the agent is in the Right square, where Suck cleans the dirty Right square (S6 → S8).
2	Percept-update step misapplied: after a percept the belief state is the set of predicted states consistent with the percept. The percept [Left, Dirty] is a pair (location, status). The student matched only 'Dirty' and ignored the location, admitting S2 and S6 whose agent is on the right.
3	Transition model misapplied: Right moves an agent in the Left square to the Right square. The student left S7 unchanged, mistaking the rule 'Right in the right square leaves the agent where it is' for a general rule.
4	Misunderstands coercion in sensorless problems: actions alone can shrink a belief state (Suck, Right, Suck forces the world into S8). A solution is a plan whose final belief state contains only goal states; no percepts are needed. Uncertainty about dirt was also mistaken for uncertainty about location.

No.	Root cause (concept or rule misapplied)
5	Confuses knowing the state (singleton belief state) with having reached the goal. The goal test on a belief state requires every member to be a goal state (S7 or S8); S6 has a dirty Right square.
6	Confuses a belief state with a physical state. A belief state is a set of physical states, so the belief-state space is the power set (2^N); one belief state per physical state is wrong.
7	—

(c) Write the corrected tables and conclusions, with justification.

(i) Task A: the corrected belief states.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S8}

(ii) Task B: the corrected belief states after each action and each percept.

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

(iii) Corrected conclusions C1 and C2, with justification.

C1	Corrected C1: After [Suck, Right, Suck] the Task A belief state is the singleton {S8} = (R, C, C), a goal state, so the plan is a guaranteed solution without any sensor. Sensorless problems CAN be solved: a solution is an action sequence that takes the initial belief state to a belief state containing only goal states (coercion); the agent never needs to observe anything. Justification: Suck on {S6, S8} gives {RESULT(S6,Suck), RESULT(S8,Suck)} = {S8, S8} = {S8}. Even the earlier two-state belief {S6, S8} was not 'unknown location' – both states have the agent on the Right.
C2	Corrected C2: The belief state {S6} is a singleton, so the agent knows it is in the Right square, the Left square is clean and the Right square is dirty. But S6 = (R, C, D) is not a goal state, so the goal has NOT been reached. The agent must perform Suck, giving {S8} = (R, C, C); only then is the goal reached. Justification: a belief state satisfies the goal only if every physical state in it is a goal state (S7 or S8); knowing the state is not the same as reaching the goal.

(d) Apply the corrected concepts.

(i) How many different belief states are possible when there are 8 physical states? Why does this make belief-state search harder than search over physical states?

(i)	With 8 physical states a belief state is any subset of them, so there are 2^8 = 256 belief states (255 non-empty). This makes belief-state search harder: the belief-state space grows exponentially (N physical states give 2^N belief states, e.g. a vacuum world with n squares has n·2^n physical states but 2^(n·2^n) belief states); each belief state is itself a set of up to N states, needing more memory, and every transition applies the action to every member of the set. Only the belief states reachable from the initial one need be explored, but the space is still far larger than the physical state space.

(ii) For the initial belief state {S1, S3, S5, S7}, apply the plan [Right, Suck], write the belief state after each action, and state whether it is a solution.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8} (S1→S2, S3→S4, S5→S6, S7→S8)
2	Suck	{S4, S8} (S2→S4, S4→S4, S6→S8, S8→S8)
	Solution? Why?	No. The final belief state {S4, S8} contains S4 = (R, D, C), whose Left square is still dirty. A solution requires every state in the final belief state to be a goal state (S7 or S8). (By contrast [Suck, Right, Suck] ends in {S8} and is a solution.)

Rubrics for Error Analysis

Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)
20	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed

30	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning
25	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified
15	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance

Criteria	**Level 1 – Beginning (1)**			
Error Identification				Unable to identify causes of errors
Root Cause Analysis				Incorrect or no corrections provided
Correction & Justification				Unable to apply concepts
Application of Concepts				Poor presentation and difficult to understand
Clarity & Presentation **10**	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure Fails to identify key errors	

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____